Markov State Modelling for the folding of HP35 villin headpiece

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In this work, we compare different feature sets and dimensionality reduction methods for the construction of Markov state models (MSMs) of the folding of the HP35 villin headpiece.

I. INTRODUCTION

A. Brief description of the biological problem, knowns and unknowns

Markov State Models (MSMs) have been used for nearly two decades to extract kinetic information from molecular dynamics simulations and provide a coarse-grained description of the conformational space of biomolecules [6]. The classical pipeline for building MSMs consists of several steps, each involving a number of choices that can affect the quality of the final model. Although procedures have standardized over the years, still the process consists of stages where more than one choice is possible and free parameters must be optimized in lack of a clear minimization objective, making the process appear somewhat "handcrafted". Recently, deep learning approaches have been proposed to automate some parts or all of the process of building MSMs - most imporantly, feature selection and dimensionality reduction stages [8][9], with the aim is improving the quality of the resulting models and reducing the human-induced bias.

B. Defining the question

In this work we review the classical pipeline of MSM construction, concentrating on the feature selection and dimensionality reduction stages. The aim is to review the classical approaches before moving on to the deep learning ones, and setting up a baseline for model comparison that can be extended to comphrehend the deep learning methods in future studies.

C. Defining the plan: translating the question on the language of the computer (brief)

We use the HP35 villin headpiece as a case study, a small protein that has been extensively studied in the literature and for which long molecular dynamics simulations are available. We compare two sets of features: contact distances and dihedrals. We compare two dimensionality reduction methods: PCA and TICA. We compare the resulting models in terms of accuracy of the folding time estimate, ability to provide a structural characterization of the metastable states, and robustness to hyperparameter choice.

II. METHODS

A. Dataset assembly

All analyses presented in this work are based on a single $300\,\mu$s molecular dynamics trajectory of the fast-folding Lys24-Nle/Lys29-Nle mutant of the 35-residue villin headpiece subdomain (HP35), simulated at 360 K. HP35 is a prototypical ultrafast-folding protein with an experimentally measured folding time of approximately $0.7,\mu$s at this temperature and has become a standard benchmark system for Markov state model construction and validation [3–5]. The trajectory was originally generated by D. E. Shaw Research on the Anton supercomputer and reported by Piana *et al.* [1]. Their simulations predicted a folding time of approximately $1.5,\mu$s, reasonably close to the experimental estimate of $0.73,\mu$s for the same mutant, although still larger by a factor of about $\times 2.5$. At the time of this work, the original trajectory generated by D. E. Shaw Research was temporarily unavailable due to maintenance of the data repository. We therefore used a preprocessed version distributed by Nagel *et al.* through their GitHub repository, which has been adopted in recent benchmark studies of Markov state modeling [3, 4]. The timeseries consists of approximately 1.5×10^6 samples, with a sampling interval of $\simeq 200\,ps$.

B. Methods used for analysis (in case you developed a method, it goes into Results)

All code is written in Python and is available on GitHub at . Markov state models were built using the Deeptime library [7]. Since MSM construction is a multi-step process, it is especially important to understand clearly the choices made at each step and validate them before moving on to the next one. The principle "garbage in, garbage out" applies here, and a mistake at an early stage can propagate and affect the final results. For this reason, in this section we describe synthetically the motivation for each step and define the methods that were investigated and compared in this work.

1. Feature selection: contact distances and dihedrals.

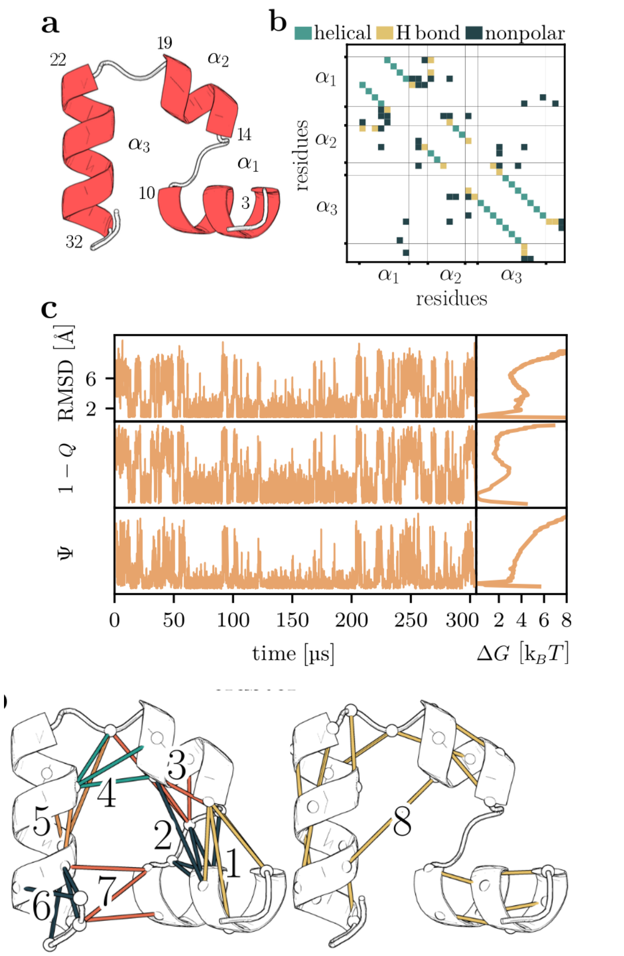

FIG. 1: *Folding of villin headpiece (HP35). (a) Molecular structure, consisting of three* α*-helices (residues* $3-10$, $14-19$ *and* $22-32$*) that are connected by two loops. (b) Contact map showing the* 42 *native contacts of HP35. (c) Time evolution of the RMSD, compared to* $1-Q$ *(with* Q *being the fraction of native contacts) and* Φ *representing the sum over the backbone dihedral angles* ϕ_i *of the three* α*-helices. The right side shows the corresponding free energy curves (in units of* k_BT*) along these coordinates.*

2. Dimensionality reduction: PCA and TICA.

3. Clustering into microstates.

4. MSM estimation: selection of lag time. Validation: implied timescales test.

5. Coarse-graining into macrostates: PCCA and MPP. Validation: CK test.

Feature definition and filtering The feature representations employed in this work were adopted from the preprocessing pipeline of Nagel *et al.* [3]. We therefore provide only a brief summary here and refer the reader to the original publication for a detailed discussion. Two classes of structural descriptors were considered: native-contact distances and backbone dihedral angles. While the total number of pairwise residue distances grows quadratically with protein size, native contacts provide a compact representation focused on interactions that are relevant to the folding process. Following Nagel *et al.*, native contacts were defined from the folded structures observed in the simulation rather than directly from the crystal structure. Distance and frequency cutoffs were applied to exclude transient contacts, yielding a set of 42 contact-distance features. Backbone conformations were described by the ϕ and ψ dihedral angles. Instead of the standard sine–cosine transformation commonly used to handle angular periodicity, we adopted the angular representation proposed by Nagel *et al.*, based on shifting the angular coordinates such that the largest gap in the Ramachandran distribution coincides with the periodic boundary. This minimizes distortions introduced by Euclidean distance measures while preserving the original angular coordinates [12]. As typically only a small subset of those coordinates is involved in a specific biomolecular process, it is important to discard the remaining uncorrelated motions or weakly correlated noise coordinates. This is because they may exhibit large amplitudes or long time scales and therefore will be erroneously considered important by PCA and TICA, respectively. To remove motions that are unlikely to contribute to the folding dynamics, the authors further applied the MoSAIC correlation analysis [3], which identifies groups of collectively varying coordinates and filters out largely uncorrelated motions. After this filtering step, the final feature sets consisted of 42 native-contact distances and 62 backbone dihedral angles, with the dihedrals ψ_1, ψ_2, ψ_{34}, ϕ_2, ϕ_{34}, and ϕ_{35} excluded from the analysis.

Dimensionality reduction and clustering The feature representations described above are high-dimensional and therefore unsuitable for direct clustering. A dimensionality-reduction step is commonly employed to obtain a compact representation of the conformational ensemble while retaining the information relevant to the slow dynamics. Ideally, conformations that rapidly interconvert should remain close in the reduced space, whereas conformations separated by slow transitions should be well separated. Since microstates are subsequently defined by clustering in this reduced space, the quality of the dimensionality reduction directly affects the quality of the resulting Markov state model. Many methods have been proposed, both linear and nonlinear, some aided by deep learning.

In this work we restrict ourselves to considering two widely used linear dimensionality-reduction methods: principal component analysis (PCA) and time-lagged independent component analysis (tICA). PCA identifies orthogonal directions of maximal variance in the data, whereas tICA identifies directions of maximal autocorrelation at a specified lag time. Because slow molecular processes are typically associated with slowly decorrelating coordinates, tICA is often regarded as more suitable for MSM construction. We compare different variations of tICA, namely i) no normalization, ii) kinetic map rescaling and ii) compute time rescaling [S1]. After dimensionality reduction, conformations were partitioned into microstates using the k-means clustering algorithm.

MSM estimation and hyperparameter selection with VAMP We estimated the transition matrix from the empirical count matrix of transitions between microstates and applying a maximum likelihood estimator. The selection of the lag time was one of the hyperparameters to optimize and its optimization was done as explained in the next section.

The number of retained components and the number of microstates constitute important hyperparameters of the MSM construction procedure.

For tICA, the relevant hyperparameters are the number of retained time-lagged independent components and the tICA lag time (τ_{tICA}). The corresponding eigenvalues quantify the autocorrelation of the projected coordinates at lag time (τ_{tICA}).

Historically, the selection of these hyperparameters relied largely on heuristic criteria and subsequent validation of the resulting MSM through Chapman-Kolmogorov (CK) tests. More recently, variational approaches such as the Variational Approach for Markov Processes (VAMP) [10, 11] have introduced objective scoring functions that can be used to compare different choices of features, dimensionality-reduction parameters, and clustering schemes. These scores provide a principled framework for model selection and are discussed in more detail in the Supporting Information.

The hyperparameters of the dimensionality-reduction and clustering pipeline were selected by comparing cross-validated VAMP-2 scores over a grid of tICA lag times, retained tICs, and numbers of k-means microstates. The model with the highest validation VAMP-2 score was selected, subject to subsequent MSM validation. Maybe also implied timescales test?

MSM validation The models selected were validated with a Chapman-Kolmogorov test [S2].

C. Statistical analysis

Here talk about everything you do after you have the validated MSM: the TPT analysis, the estimation of pathways, and the coarse graining.

III. RESULTS

This is the real talk, the biological talk! Comparative analysis of the results of the various MSM constructed.

Sections describing different aspects of your analysis (3a-3d). All sections should be organised as follows, the text is continuous, broken only to a few paragraphs in each section:

- Question studied in the section and the goal of this calculation.
- Few sentences on how it was done, mostly referring to datasets and approaches in the Methods.
- Actual results and observations, each supported by either a i) table or ii) figure or iii) number in the text.
- You need to interpret the observation. For example, do not only write X ¿ Y, but also the meaning of this relationship with respect to your question.
- When presenting the results, refer to the respective Figure and Table, where it was shown.
- You can add more sentences for the interpretation. Please use precise phrases for the level of evidence, if the results 'suggest', 'indicate', 'show','demonstrate', 'support', 'corroborate'. Do not put too much 'hypothesis', and 'propose' these are synthetic phrases.
- End each section with a conclusion sentence for example, 'Taken together the results consistently show that X is larger than Y, indicating that X was an ancestor of Y'.

IV. DISCUSSION

Here you try to put your results into a broader context and present your scientific view/model based on that. This is a continuous text, broken into paragraphs. In case you have two distinct 'huge' conclusions, you may generate separate sections (max 2). Chapter 4 should be organised as follows:

- You may phrase your biological question again.
- Briefly summarise the results (max one paragraph). You do not repeat the words you used earlier, make it concise. Make it evident how your results answer the question you have asked.
- Compare your results with something in the literature (use PubMed, or literature indicated in UniProt, PDB or any datasets we learnt about). 1-2 articles enough. The point is that you

- show how your results relate to previous ones. Different scenarios: 'support', 'in contrast', 'in addition', 'gives insights into'.

- If possible, present a model. This can be a molecular 'mechanism', 'scenario', 'reason'. For example: 'A and B belong to the same family'; 'mutation XD likely changes the interaction in liver'; 'in disease T we have less phosphorylation sites or the level of phosphorylation decreased'; 'in this experiment, there is a problem with degradation of protein P. ' etc.

- The model can be presented as: 'propose', 'suggest', 'indicate','hypothesize', or even 'support' or 'demonstrate'. If possible try to present a cartoon figure or scheme.

V. CONCLUSIONS

One paragraph, emphasizing the main finding and formulate a message for the future (perspective), how to continue from here, what kind of further studies are needed. You may have Conclusion as chapter 5.

VI. REFERENCES

The text should contain references, to previous results, data, methods, softwares, etc.. Format is unrestricted, but you should use these refs in the text, and list them at the end. I expect ca 15-20 refs listed.

[1] S. Piana, K. Lindorff-Larsen and D.E. Shaw, *Protein folding kinetics and thermodynamics from atomistic simulation*, Proc. Natl. Acad. Sci. U.S.A. 109 (44) 17845-17850 (2012), DOI: 10.1073/pnas.1201811109.

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[11] H. Wu and F. Noé, *Variational approach for learning markov processes from time series data*, Journal of Nonlinear Science 30(1):23-66, 2020. DOI: 10.1007/s00332-019-09576-8.

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[14] C.R. Schwantes and V.S. Pande, *Improvements in Markov State Model Construction Reveal Many Non-Native Interactions in the Folding of NTL9* J. Chem. Theory Comput. 2013, 9, 4, 2000-2009 DOI: 10.1021/ct300878a.

SUPPORTING INFORMATION

S1. Dimensionality reduction and clustering

PCA and tICA are both linear dimensionality reduction techniques, i.e. they project the data onto a linear subspace of the original feature space. Let $\{\mathbf{x}(t)\}_{t=0}^{T-1}$ be a multidimensional, discrete time-series, where each snapshot is a column vector, of dimension D, corresponding to an arbitrary, vectorized representation of a protein conformation (e.g. a set of dihedral angles). Suppose the data has been sampled from a *stationary* and *time-reversible* stochastic process. Without loss of generality, we assume that the data have been centered, $\mathbb{E}[\mathbf{x}] = 0$. Using the braket notation $\langle x| := \mathbf{x}^T, |x\rangle := \mathbf{x}$, we define the covariance matrix:

$$\mathbf{C}_0 := \mathbb{E}[|x\rangle \langle x|] \tag{1}$$

and the time-lagged covariance matrix:

$$\mathbf{C}_{0,\tau} := \mathbb{E}\left[|x(t)\rangle \langle x(t+\tau)|\right]. \tag{2}$$

For a stationary reversible process,

$$\mathbf{C}_{0,\tau} = \mathbf{C}_{\tau,0}^T = \mathbf{C}_{\tau,0},$$

so that the lagged covariance matrix may be denoted simply by $\mathbf{C}_\tau$.

If $|\hat{u}\rangle \langle \hat{u}|$ is a projection operator (i.e. $|\hat{u}\rangle$ is a unit vector), then:

$$\langle \hat{u}| \mathbf{C}_0 |\hat{u}\rangle = \mathbb{E}[\langle \hat{u}|x\rangle \langle x|\hat{u}\rangle] = \mathbb{E}[\langle \hat{u}|x\rangle^2] \tag{3}$$

is the variance of the projection of the data along $|\hat{u}\rangle$, and

$$\begin{aligned} \mathrm{ACF}_{\hat{u}}(\tau) &= \frac{\langle \hat{u}| \mathbf{C}_\tau |\hat{u}\rangle}{\langle \hat{u}| \mathbf{C}_0 |\hat{u}\rangle} \\ &= \frac{\mathbb{E}[\langle \hat{u}|x(t)\rangle \langle \hat{u}|x(t+\tau)\rangle]}{\mathbb{E}[\langle \hat{u}|x\rangle^2]} \end{aligned} \tag{4}$$

its autocorrelation at lag time τ.

PCA seeks the direction of maximal variance. The first principal component is defined as

$$\begin{aligned} |\hat{u}_1\rangle = &\arg\max \langle \hat{u}| \mathbf{C}_0 |\hat{u}\rangle \\ &\text{subject to } \|\hat{u}\| = 1 \end{aligned} \tag{5}$$

this optimization problem can be solved introducing a Lagrange multiplier λ, and the stationarity condition yields

$$\mathbf{C}_0 |\hat{u}_i\rangle = \lambda_i |\hat{u}_i\rangle, \tag{6}$$

i.e. the projection versor is an eigenvector of the covariance matrix. The corresponding eigenvalue equals the variance of the data projected onto the principal component:

$$\langle \hat{u}_i| \mathbf{C}_0 |\hat{u}_i\rangle = \lambda_i \cdot \langle \hat{u}_i|\hat{u}_i\rangle = \lambda_i.$$

Then the second principal component is defined as the direction of maximal variance orthogonal to the first, giving the optimization problem:

$$\begin{aligned} |\hat{u}_2\rangle = &\mathrm{argmax} \langle u| \mathbf{C}_0 |u\rangle \\ &\text{subject to } \langle u|u\rangle = 1 \\ &\text{and } \langle u|\hat{u}_1\rangle = 0 \end{aligned} \tag{7}$$

which again can be solved with Lagrange multipliers. Iterating, this implies that the principal components form an orthonormal basis of eigenvectors of the covariance matrix, i.e. they diagonalize the covariance matrix. Dimensionality reduction ($d < D$) is achieved by taking the components along the first d principal components $\{|\hat{u}_1\rangle, \ldots, |\hat{u}_d\rangle\}$.

$$z_i = \langle \hat{u}_i|x\rangle, \qquad \mathbf{z} = U^T\mathbf{x}. \tag{8}$$

This is a linear projection onto an ortonormal basis:

$$|x\rangle \to |z\rangle := \sum_{i=1}^{d} \langle x|\hat{u}_i\rangle \, |\hat{u}_i\rangle$$

By construction, in the reduced space components are uncorrelated and ordered by decreasing variance:

$$\mathbb{E}[|z\rangle \langle z|] = \mathrm{diag}(\lambda_1, \cdots \lambda_d) \tag{9}$$

tICA Unlike PCA, tICA [14] seeks directions $|u\rangle$ that maximize the autocorrelation Eq.(4) at a fixed lag time τ. The lagtime τ is a free parameter. The autocorrelation is invariant under scaling of the projection vector $|u\rangle$. Differently from PCA, the normalization constraint is usually implemented as $\langle u| \mathbf{C}_0 |u\rangle = 1$, as this choice simplifies the optimization problem. Formally, tICA solves the optimization problem:

$$\begin{aligned} |\hat{u}_1\rangle = &\mathrm{argmax} \langle u| \mathbf{C}_\tau |u\rangle \\ &\text{subject to } \langle u| \mathbf{C}_0 |u\rangle = 1 \end{aligned} \tag{10}$$

using the same procedure as for PCA, we can solve this with Lagrange multipliers, and show that

$$\mathbf{C}_\tau \left|\hat{u}_1\right\rangle = \lambda_1 \mathbf{C}_0 \left|\hat{u}_1\right\rangle, \tag{11}$$

.

The generalized eigenvalue λ_i is equal to the autocorrelation of the projected coordinate $\langle \hat{u}_i | x(t)\rangle$ at lag time τ. Assuming that this coordinate approximates a single relaxation mode of the dynamics, its autocorrelation decays exponentially, yielding

$$\lambda_i \approx e^{-\tau/t_i}, \tag{12}$$

where t_i is the corresponding relaxation timescale. In practice, the relation is approximate because a tIC may contain contributions from multiple dynamical modes.

Similarly as done with PCA, we can iterate this procedure and define the second tIC as the direction of maximal autocorrelation at lag time τ that is *uncorrelated* with the first:

$$\begin{aligned} \left|\hat{u}_2\right\rangle = &\text{argmax} \left\langle u\right| \mathbf{C}_\tau \left|u\right\rangle \\ &\text{subject to } \left\langle u\right| \mathbf{C}_0 \left|u\right\rangle = 1 \\ &\text{and } \left\langle u\right| \mathbf{C}_0 \left|\hat{u}_1\right\rangle = 0 \end{aligned} \tag{13}$$

the solution $|\hat{u}_1\rangle$ is again a solution of the same generalized eigenvalue problem as in Eq.(10). generalized eigenvalue problem. Dimensionality reduction ($d < D$) is achieved by taking the components of the data $\mathbf{x}(t)$ along the first d tICs $\{|\hat{u}_1\rangle, \ldots, |\hat{u}_d\rangle\}$.

$$\mathbf{z}(t) = (\langle x(t)|\hat{u}_i\rangle, \cdots, \langle x(t)|\hat{u}_d\rangle) \tag{14}$$

By construction, in the reduced space components z_i have unit variance, are uncorrelated at zero lag, and are ordered by decreasing autocorrelation at lag time τ.

$$\mathbb{E}[|z\rangle\langle z|] = \mathbb{I} \tag{15}$$

tICA as a diagonalization in the whitened coordinate space It is worth noting that unlike in PCA, the tIC vectors are generally not orthogonal under the standard Euclidean inner product. Instead, as we said, they satisfy the generalized orthogonality relation

$$\left\langle u_i\right| \mathbf{C}_0 \left|u_j\right\rangle = \delta_{ij}.$$

i.e. they corresponding components z_i and z_j are *uncorrelated* at zero lag. Nevertheless, the tICA procedure can be reformulated as a standard eigenvalue problem in a transformed coordinate system. The key idea is to first apply a whitening transformation, which rescales and rotates the original data so that all directions have unit variance and are mutually uncorrelated. Define the ZCA whitening transformation

$$|y\rangle = \mathbf{C}_0^{-1/2} |x\rangle$$

where $C_0^{-1/2}$ is symmetric and positive definite. By construction, the covariance matrix in the transformed space is the identity:

$$\mathbb{E}\left[|y\rangle\langle y|\right] = \mathbf{C}_0^{-1/2}\mathbf{C}_0\mathbf{C}_0^{-1/2} = \mathbf{I}.$$

and time-lagged covariance matrix in the whitened space is

$$\hat{\mathbf{C}}_\tau = \mathbf{C}_0^{-1/2}\mathbf{C}_\tau\mathbf{C}_0^{-1/2}$$

$\hat{\mathbf{C}}_\tau$ is symmetric and can thus be diagonalized:,

$$\hat{\mathbf{C}}_\tau |v_i\rangle = \lambda_i |v_i\rangle, \qquad \langle v_i|v_j\rangle = \delta_{ij}.$$

If $|u_i\rangle$ solves the tICA problem (10), then $|v_i\rangle = \mathbf{C}_0^{1/2}|u_i\rangle$ is an eigenvector of $\hat{\mathbf{C}}_\tau$ with the same eigenvalue.

Indeed, multiplying the generalized eigenvalue problem (10) from the left by $\mathbf{C}_0^{-1/2}$ and defining $|v_i\rangle = \mathbf{C}_0^{1/2}|u_i\rangle$ yields

$$\hat{\mathbf{C}}_\tau |v_i\rangle = \lambda_i |v_i\rangle.$$

Consequently, the tICA components z_i correspond to the ortogonal projection of the whitened data $\mathbf{y}$ onto the ortonormal basis defined by $\{|v_i\rangle\}$:

$$z_i(t) = \langle x(t)|u_i\rangle = \langle x(t)|C_0^{-1/2}v_i\rangle = \langle y(t)|v_i\rangle.$$

Alternative generalized eigenvector normalizations for tICA
The construction presented above corresponds to the original formulation of tICA, in which the covariance matrix of the transformed data is the identity (15). However, alternative rescaling of the transformed coordinates have been used in the literature and are usually implemented in softwares to endow Euclidean distances in the reduced space with a more direct kinetic interpretation. These can be particularly useful when the tICA coordinates are subsequently clustered to construct a Markov state model, as we do in this work.

The most commonly used choice is the *kinetic map* in which each tIC component z_i is multiplied by its corresponding eigenvalue,

$$\tilde{z}_i(t) = \lambda_i z_i(t). \tag{16}$$

With this normalization, the coordinates representing the slowest processes "weight more", and a unit step in those directions in the reduced space is assigned a larger distance. An alternative is the *commute map*, where each coordinate is rescaled according to its implied timescale,

$$\tilde{z}_i(t) = \sqrt{\frac{t_i}{2}} z_i(t), \qquad t_i = -\frac{\tau}{\ln \lambda_i}. \tag{17}$$

With this normalization, Euclidean distances approximate commute distances, i.e. the expected time required to travel from one configuration to another and back. This is a variation on the same idea behind the kinetic map, being eigenvalues and timescales connected by (12).

S2. Estimation and validationof Markov state models

CK test

To insert.

S3. VAMP scores

S4. Transition path theory

S5. PCCA and MSM coarse-graining

SUPPLEMENTARY FIGURES

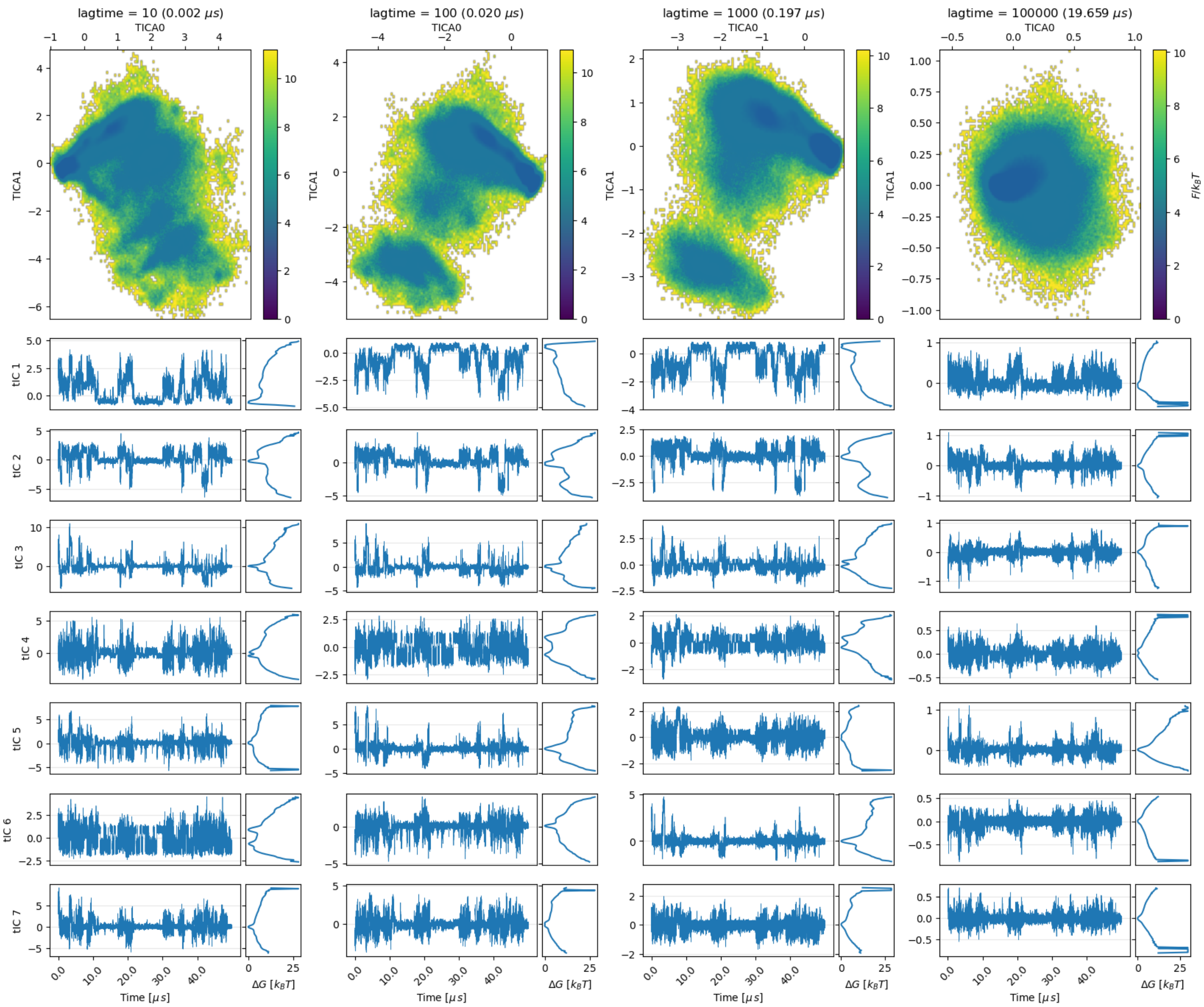

FIG. SF1: tICA on dihedrals feature set. "Kinetic map" scaling is used.

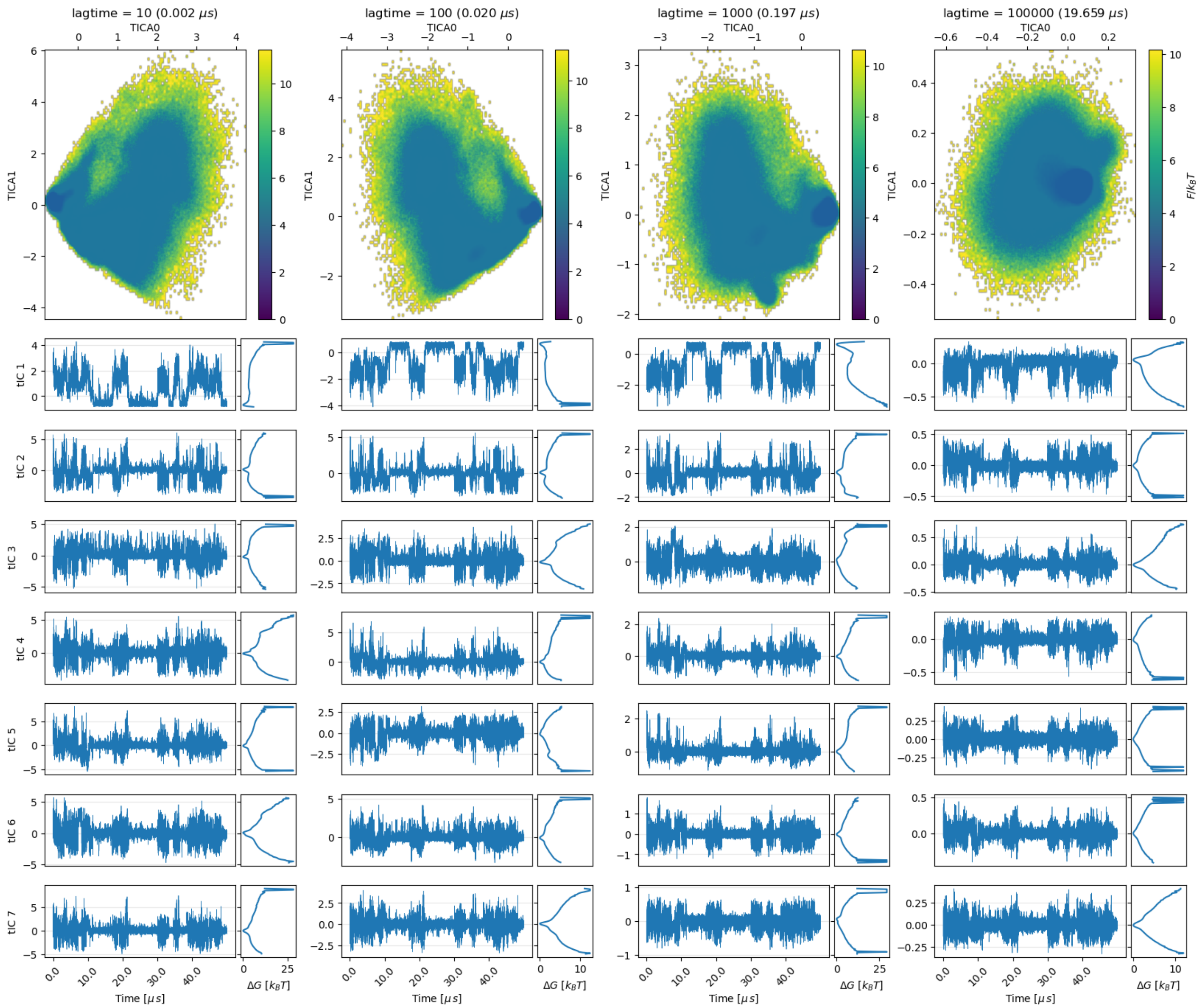

FIG. SF2: tICA on distances feature set. ”Kinetic map” scaling is used.

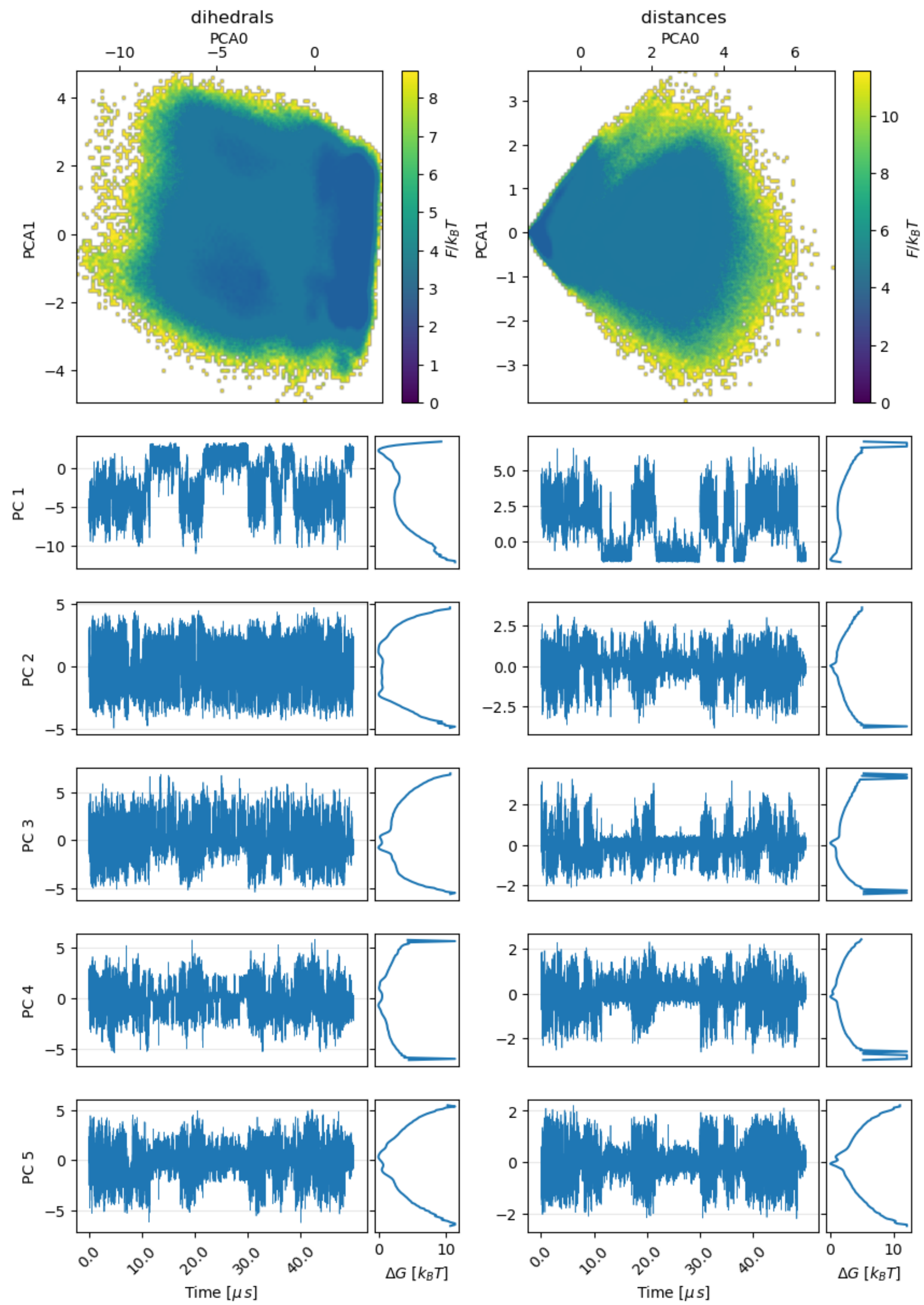

FIG. SF3: PCA on free energy landscape.